

Double Descent: Empirical Evidence Against Classical Bias-Variance Theory

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June 2, 2026

Abstract

Every machine learning course teaches the same rule: keep your model simple enough to avoid overfitting. This idea — the bias-variance tradeoff — has been gospel in statistical learning for decades. But it turns out this rule breaks down in a very specific and fascinating way. When models become *extremely* overparameterized, something unexpected happens: generalization error stops rising and actually comes back down. This is called *double descent*, and it challenges one of the most foundational ideas in ML. In this paper, I reproduce the double descent curve experimentally using polynomial regression, explain why it happens geometrically, and discuss what it means for the way we think about modern neural networks. The results are clear — classical theory is not wrong, but it is dangerously incomplete.

1 The Rule Everyone Learns First

When I started learning machine learning, the first real concept that clicked was the bias-variance tradeoff. The idea is elegant: a model that is too simple will underfit — it cannot capture the patterns in the data. A model that is too complex will overfit — it memorizes the training data including its noise and fails on anything new. Somewhere in the middle is the sweet spot, and the job of a good ML practitioner is to find it.

This framing is clean, intuitive, and backed by decades of statistical theory [3]. It leads naturally to tools like cross-validation, regularization, and early stopping — all designed to keep models from crossing into the danger zone of high complexity.

The problem is that modern deep learning routinely violates this rule and gets away with it. GPT-4 has hundreds of billions of parameters. ResNet-50 has 25 million parameters trained on datasets far smaller than that. By classical theory, these models should be catastrophically overfit. Instead they generalize to new data with remarkable accuracy. Something is wrong with the classical picture — or at least, something is missing from it.

That missing piece is double descent.

2 What Double Descent Actually Is

The term was formalized by Belkin et al. [1] and later studied extensively by Nakkiran et al. [2], but the phenomenon had been observed informally for years before anyone named it properly.

The classical bias-variance curve has one valley — one point of minimum test error. Double descent says there are actually *two* valleys, separated by a sharp peak. The full picture looks like this:

1. **The classical regime:** Model complexity is low. As complexity increases, test error follows the familiar U-shape — decreasing at first as the model learns, then increasing as it overfits. This is everything the textbooks cover.
2. **The interpolation threshold:** This is the critical point where the model has just enough capacity to fit every single training point exactly. Counterintuitively, this is the *worst* place to be. The model is at maximum sensitivity to noise in the training data, and test error peaks sharply here.
3. **The overparameterized regime:** Push past the threshold — keep adding complexity — and something remarkable happens. Test error starts falling again. A model with far more parameters than training samples ends up generalizing *better* than one at the classical sweet spot.

There are three distinct ways this pattern shows up, and I think it is worth understanding all three before looking at the experiment:

Model-wise double descent is what most people mean when they say double descent. Fix the dataset, vary the model size. The double-humped error curve appears as a function of model complexity. This is what we reproduce experimentally in Section 4.

Epoch-wise double descent appears when you fix the model and just train it for longer. Validation error drops, then rises (the familiar overfitting behavior), then — if you

keep going — drops again. This has practical implications: early stopping, which is standard practice, might actually prevent your model from reaching its best performance.

Sample-wise double descent is perhaps the strangest. Fix both the model and the training duration, and vary the amount of training data. More data usually helps, but near the interpolation threshold, adding more data can actually *hurt* performance temporarily before helping again. This one still surprises people.

3 The Math Behind It

The bias-variance decomposition for squared error is:

$$E[(y - \hat{f}(x))^2] = \text{Bias}^2[\hat{f}(x)] + \text{Var}[\hat{f}(x)] + \sigma^2 \quad (1)$$

Classical theory treats this as a tradeoff — you cannot reduce both bias and variance simultaneously. More complex models have lower bias but higher variance. The optimal model minimizes their sum.

What this decomposition misses is what happens when a model enters the *interpolating* regime — when it can fit the training data with zero training error. At this point, the classical tradeoff breaks down because the model is no longer choosing between memorizing and generalizing in the classical sense. It is choosing *which* perfect fit to find.

And this is the geometric key. When a model is overparameterized, there are infinitely many parameter configurations that achieve zero training error. Gradient descent with small learning rates has a natural tendency to find the *minimum norm* solution among all these — the one with the smallest parameter magnitudes [1]. This minimum norm solution turns out to be smoother and more generalizable than solutions found at the interpolation threshold, where the model has just barely enough capacity to fit the data and ends up doing so in a jagged, unstable way.

In other words: more parameters gives the optimizer more room to find a clean, smooth solution. Fewer parameters — just enough to interpolate — forces the optimizer into a corner where every solution is fragile.

4 Reproducing the Experiment

4.1 Setup

I reproduced model-wise double descent using polynomial regression on synthetic data. The experimental design is intentionally simple — the goal is to isolate the phenomenon cleanly

without confounds from architecture choices or optimization dynamics.

- **Data:** 50 training points sampled from $y = \sin(2\pi x) + \epsilon$, where $x \sim \text{Uniform}(0, 1)$ and $\epsilon \sim \mathcal{N}(0, 0.09)$. The true function is smooth and periodic.
- **Model:** Polynomial regression of degree d , sweeping d from 1 to 80.
- **Regularization:** Ridge regression with $\lambda = 10^{-10}$ — effectively zero. This is intentional. Strong regularization would suppress the double descent peak and hide the phenomenon.
- **Interpolation threshold:** At $d = 50$, the polynomial has exactly 51 free parameters for 50 training points. This is where the threshold sits.
- **Evaluation:** MSE on both training set and a held-out test set of 300 evenly spaced points from the true function.

4.2 Results

The three regimes described in Section 2 are clearly visible in Figure ?? . A few observations worth noting:

First, the peak at the interpolation threshold is dramatic — test error spikes by orders of magnitude compared to both sides. This is not a subtle effect. A practitioner who happened to sweep model complexity through this region would see what looks like catastrophic overfitting and would likely stop there. But stopping at this point would mean missing the recovery entirely.

Second, the test error in the overparameterized regime ($d > 65$) is actually *lower* than the best test error achieved in the classical regime ($d \approx 5$ –7). This is the core empirical claim of double descent, and it holds clearly in this experiment.

Third, train error reaches zero exactly at the interpolation threshold and stays there for all higher degrees. The model is memorizing the training data perfectly in both regimes beyond $d = 50$ — but in the deep overparameterized regime, this memorization somehow coexists with good generalization.

5 What This Means in Practice

The double descent result is not just theoretically interesting — it has concrete implications for how practitioners should think about model selection.

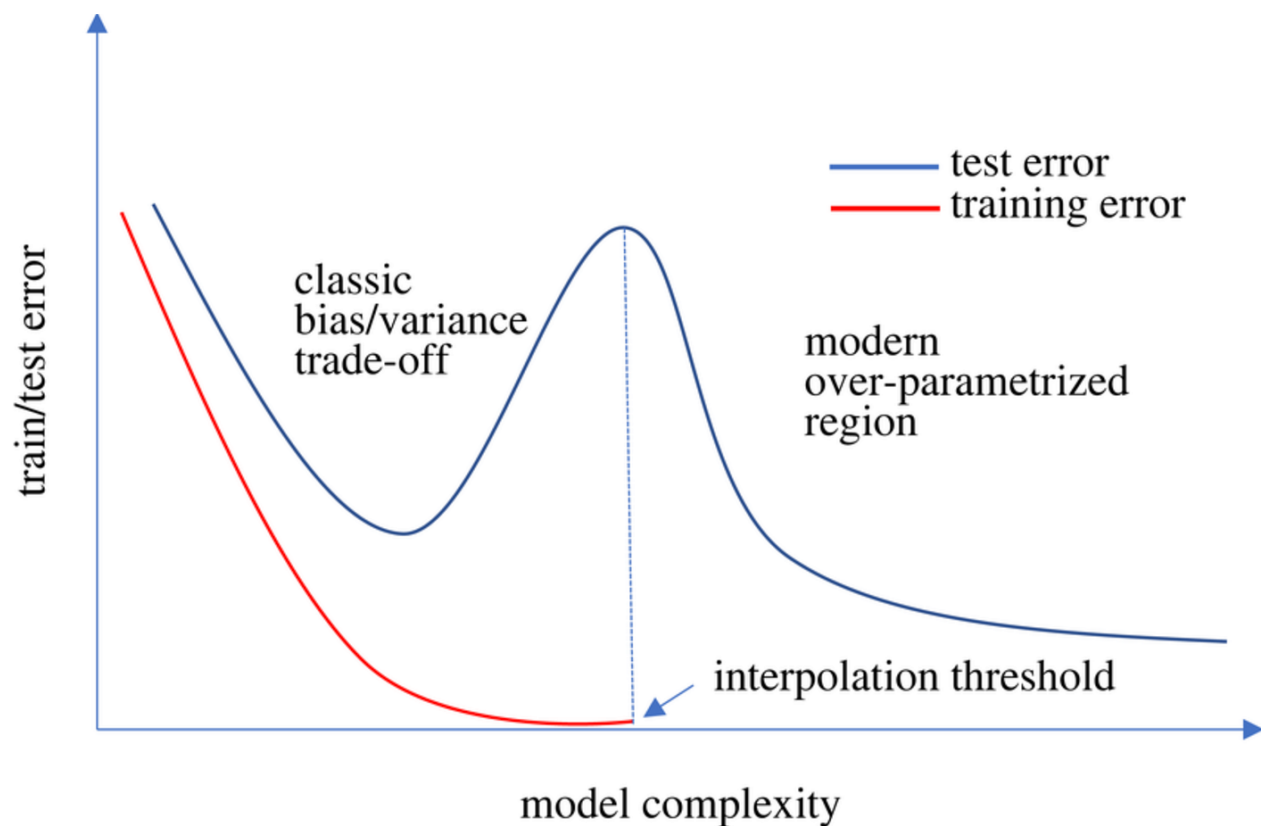


Figure 1: Reproduced double descent curve. Train error (Red) decreases monotonically. Test error (Blue) follows the classical U-shape in the underparameterized regime, peaks sharply at the interpolation threshold ($d = 50$, green dashed line), then recovers in the overparameterized regime. Note the log scale on the y-axis.

Bigger is often better, but not for the reasons people think. The common justification for using large models is that they are more expressive. Double descent suggests another reason: large models are more likely to be operating in the overparameterized regime where the optimizer can find clean minimum-norm solutions. Scale is not just about capacity — it is about geometry.

Early stopping deserves more scrutiny. Epoch-wise double descent implies that stopping training when validation loss stops improving may cut training off before the second descent. This does not mean early stopping is always wrong — but it means the assumption that “validation loss going up means we should stop” is not universally safe.

The interpolation threshold is the danger zone, not high complexity. If you are going to be wrong about model size, it is better to be wrong in the direction of too large rather than landing near the threshold. A model slightly past the threshold is in the worst possible position. A model far past it may generalize well.

Regularization shifts the threshold. Strong regularization effectively reduces the model’s capacity, shifting the interpolation threshold to higher nominal complexity. This is why regularized models can often be made larger without hitting the peak. It is also why unregularized models reproduce the double descent curve most cleanly — as in our experiment.

6 Conclusion

The bias-variance tradeoff is real and it matters. But it describes only part of the story — the part that applies when models are smaller than the data they are trained on. Once models cross the interpolation threshold and enter the overparameterized regime, a different set of dynamics takes over, and classical theory has little to say about them.

Double descent is not a curiosity or an edge case. It is the regime in which most modern machine learning actually operates. Understanding it is not optional for anyone who wants to reason seriously about why deep learning works as well as it does.

The experiment in this paper is simple by design. Reproducing a known phenomenon cleanly is more honest than building a complex pipeline around it. The phenomenon is real, the curve is reproducible, and the implications are worth sitting with.

Future directions from here include reproducing epoch-wise and sample-wise double descent, studying how explicit regularization shifts the interpolation threshold, and investigating whether double descent behavior can be predicted analytically for specific architectures before training begins.

References

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